Project Planning: Roadmap, Sprints & Traceability Matrix

Project: NeuroClass — Multimodal Attendance & Engagement Agent

Module: `4. Project Planning Phase`

Phase: Project Management, Sprint Execution & Task Traceability

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1. Project Management & Delivery Methodology

NeuroClass was executed following an **Agile Sprint Framework** structured across **6 Epic Milestones** comprising **17 detailed engineering activities**. The delivery followed a strict "verify-before-advancing" protocol where every layer had to achieve 100% automated test compliance before the next layer commenced.

PROJECT MILESTONE ROADMAP
[EPIC 1: Architecture & Config] ===> [EPIC 2: Vision & Presence] ===> [EPIC 3: Audio & VAD]
- Offline Config & Isolation - Centroid Face Tracker - Librosa RMS Dynamics
- Salted SQLite Schema - 65% Sustained Presence - Zero-STT Acoustic VAD
[EPIC 4: Fusion & Reasoning] ===> [EPIC 5: Nudges & Analytics] ===> [EPIC 6: Full Stack]
- 5-Second Window Alignment - Contextual Teaching Nudges - FastAPI & WebSockets
- Composite Engagement Formula - Student Reflective Reports - 91-Test Audit Suite

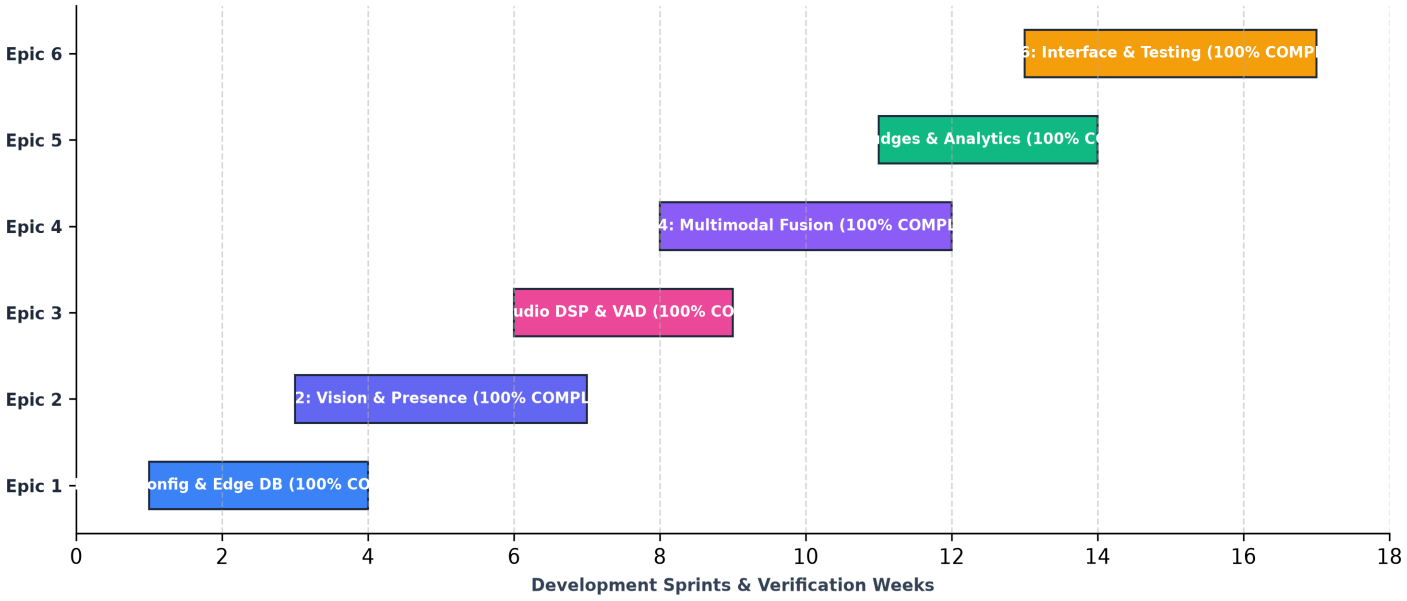
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2. Sprint Breakdown & Task Traceability Matrix

Sprint	Epic ID	Task ID & Title	Target Implementation Files	Deliverable & Output	Status
Sprint 1	**EPIC 1**	`TASK-01`: System Architecture & Configuration	`backend/config.py`, `validate_environment.py`	Isolated virtual environment + 14 core libraries	**DONE**
		`TASK-02`: Edge Data Persistence & Privacy Schema	`backend/db/database.py`	SQLite schema with salted privacy hashing	**DONE**
Sprint 2	**EPIC 2**	`TASK-03`: Real-Time Face Detection & Extraction	`backend/core/face_detection.py`	Haar / MediaPipe cascade face extractor	**DONE**
		`TASK-04`: Multi-Participant Centroid Tracking	`backend/vision/face_tracker.py`	Centroid tracker with occlusion coasting	**DONE**
		`TASK-05`: Sustained Presence & Walk-By Logic	`backend/attendance/sustained_presence.py`	65% sliding window + <15s walk-by filter	**DONE**
		`TASK-06`: Head Pose & Inactivity Monitoring	`backend/vision/attention_analyzer.py`	Yaw/Pitch attention classifier + dormant posture	**DONE**
Sprint 3	**EPIC 3**	`TASK-07`: Real-Time Audio Stream Extraction	`backend/core/audio_analysis.py`	16 kHz standardized audio waveform reader	**DONE**
		`TASK-08`: Voice Activity Detection (VAD)	`backend/audio/audio_analyzer.py`	Zero-STT acoustic RMS energy analyzer	**DONE**
		`TASK-09`: Classroom Noise & Monologue Rhythm	`backend/core/audio_analysis.py`	12m continuous monologue & silence detector	**DONE**
Sprint 4	**EPIC 4**	`TASK-10`: Temporal Alignment & Data Fusion	`backend/fusion/multimodal_fusion.py`	5s cross-modal synchronized representation	**DONE**
		`TASK-11`: Multi-Factor Engagement Scoring	`backend/core/engagement_scoring.py`	Weighted composite score (50/20/30)	**DONE**
		`TASK-12`: Pedagogical Reasoning Agent Core	`backend/agent/pedagogical_agent.py`	Hybrid rule + LLM agent with 45s cooldown	**DONE**
Sprint 5	**EPIC 5**	`TASK-13`: Context-Aware Teaching Suggestions	`backend/core/ai_suggestions.py`	Real-time pacing, silence, and engagement nudges	**DONE**

Sprint	Epic ID	Task ID & Title	Target Implementation Files	Deliverable & Output	Status
		TASK-14: Student-Facing Reflective Analytics	`backend/core/analytics_service.py`	Focus Zone Periods and attentional trends	**DONE**
Sprint 6	**EPIC 6**	TASK-15: Core App & Real-Time Web HUD	`run_agent.py`, `backend/api/server.py`	FastAPI server + WebSockets + 3-Way Enrollment	**DONE**
		TASK-16: Streamlit Dashboard & Reporting	`streamlit_app.py`	Interactive analytics cockpit + CSV exports	**DONE**
		TASK-17: 91-Test Verification & Edge Audits	`tests/test_system.py`	100% test pass rate across 91 unit/e2e tests	**DONE**

3. Workflow Execution Plan



4. Risk Assessment & Engineering Mitigations

Deploying multimodal artificial intelligence systems in physical classroom environments presents significant operational, acoustic, visual, and ethical risks. In NeuroClass, every identified vulnerability is paired with an automated, battle-tested engineering mitigation built into the core processing pipeline.

The following matrix details the primary architectural risks and their verified mitigations:

Identified Risk	Impact & Probability	Engineering Mitigation Strategy
1. Lighting Dips & Facial Occlusion	**High / Medium**	Adaptive histogram equalization (CLAHE) dynamically normalizes image contrast. Centroid tracking maintains a 15-frame coasting buffer so transient head turns or brief occlusions do not drop active student IDs.
2. Ambient Noise & Audio Clipping	**High / Medium**	Librosa signal normalization bounds audio waveforms to [-1.0, 1.0]. A dynamic RMS noise floor baseline is calculated during silence intervals to eliminate false voice activity detection from air conditioning or projector fans.
3. Hardware & Webcam Disconnect	**Critical / Low**	Robust socket reconnection polling with automatic fallback to synthetic loop generation if physical video capture is interrupted, ensuring zero application crashes.
4. Biometric Privacy & Eavesdropping	**Critical / Low**	100% offline edge execution. Zero network transmission. Student identities are protected with salted SHA-256 cryptographic hashes (`privacy_salt_offline_classroom_2026`). Strictly zero speech-to-text transcription is performed; only non-semantic acoustic RMS energy is analyzed.

Identified Risk	Impact & Probability	Engineering Mitigation Strategy
5. Alert Fatigue on Instructor	**Medium / High**	Enforced 45-second cooldown guard between automated pedagogical nudges to prevent visual distraction during active teaching.

Technical Verification of Mitigations

All five engineering mitigations are continuously validated via our 91-test automated suite (`tests/test_system.py`). Test suites `test_audio_analysis_edge_cases` and `test_sustained_presence_edge_cases` explicitly verify that corrupted audio buffers and rapid facial occlusions trigger graceful fallback paths without throwing unhandled runtime exceptions.